Tanmay Goel

goeltanmay442@gmail.com | github.com/tanmay442 | linkedin.com/in/goeltanmay442

Full-Stack & AI Engineer — shipping web products, data pipelines and AI agents end to end.

Skills

Languages : TypeScript, JavaScript, Python, SQL, HTML/CSS, Rust*
Frontend : React 19, Next.js 16, Astro, Tailwind CSS, TanStack Query, Zustand, Recharts
Backend & Storage : Node.js, Express 5, Hono, FastAPI, PostgreSQL (Neon, pgvector), MongoDB, TimescaleDB, Redis
LLM & Agents : Vercel AI SDK, smolagents, Anthropic managed agents, LangChain, tool-calling, RAG
Edge, Cloud & Tooling : Cloudflare Workers, Vercel, Docker, GCP, AWS*, Git, GitHub Actions, Vitest, Linux, Bash

Education

Harcourt Butler Technical University (HBTU) — *Kanpur, India*
Integrated Master's in Mathematics & Data Science — *Expected May 2030*

Experience

Full-Stack Developer | *Philosophy Spread* — Oct 2025 – Present

- Architected and shipped a **neo-brutalist** community platform from scratch on Astro SSR + Cloudflare Workers, avoiding bloated templates to keep client-side JavaScript near zero
- Held a **100 Desktop / 99 Mobile PageSpeed** score by server-rendering every route, never hydrating the page
- Implemented Clerk auth with **custom Google SSO** and role-gated community features; grew the platform to **400+ active users** and 600+ signups organically
- Owned the edge rollout — Wrangler CLI, environment-variable and **DNS/nameserver** management, and GitHub Actions CI/CD — from empty repo to live on Cloudflare

Data Engineer | *Taleoid Page — Chennai, India — Remote* — Jan 2026 – Mar 2026

- Replaced manual Excel tracking with **Python and Oracle SQL** pipelines for **380+ active users** (700 total accounts)
- Built **demand and popularity scoring** over purchase history to rank new titles and surface borrowing demand
- Engineered automated **email notification** pipelines for return reminders, demand alerts and eBook licence expiry

Projects

Destr | *Next.js 16, Vercel AI SDK, Drizzle, pgvector, Clerk* — github.com/tanmay442/Destr

- Built a **serverless** tool-calling RAG agent on Next.js 16 + Vercel AI SDK v6, running on **Neon Postgres** and Clerk
- Owned the **chunking and retrieval layer**: **5 pluggable chunking strategies** (semantic, parent-child, and more) with configurable retrieval — vector vs. agentic mode, hybrid search, cross-encoder rerank
- Implemented **agentic retrieval**: an empty result triggers a rewrite-and-retry pass before an out-of-domain wall, with **grounded PDF citations**, hallucination checks and automatic ticket escalation
- Built a **golden-dataset eval harness** with a 0.7 faithfulness gate, **weekly CI** runs, a **daily cron**, and a **Redis** cache

FinSmolAgent | *FastAPI, smolagents, TimescaleDB, WebSocket, React* — github.com/tanmay442/FinSmolAgent

- Architected a **six-stage multi-agent pipeline** across **5 agents** and 22 skills, KYC screening to valuation
- Enforced a **three-tier trust boundary** with **JSON-Schema-validated handoffs**, provenance on every metric, and a **compliance approval gate** that blocks execution until a human decision
- Ran backtests in a **sandboxed worker**: workspace jail, timeout kill, deterministic seeds, package whitelist

Manufacturing SalesOps | *React 19, Express 5, MongoDB, Pusher, PDFKit* — github.com/tanmay442/BDA_Module

- Engineered a single-tenant operations platform with **RBAC** for Admins, Managers and BDAs, authorizing every route server-side from Clerk JWT session tokens
- Added real-time lead-pipeline updates through **Pusher WebSocket** channels with TanStack React Query cache invalidation, removing API polling across concurrent sessions
- Built a server-side **quotation engine** with PDFKit and auto-incrementing versions, plus a drag-and-drop Kanban pipeline and role-specific Recharts dashboards over 9 backend modules and 18 test suites

Achievements

- **1st place, Smart India Hackathon (SIH) 2026 — HBTU internal college hackathon round**
- **Philosophy Spread — Logic Module 04**: 160k+ unique page visits